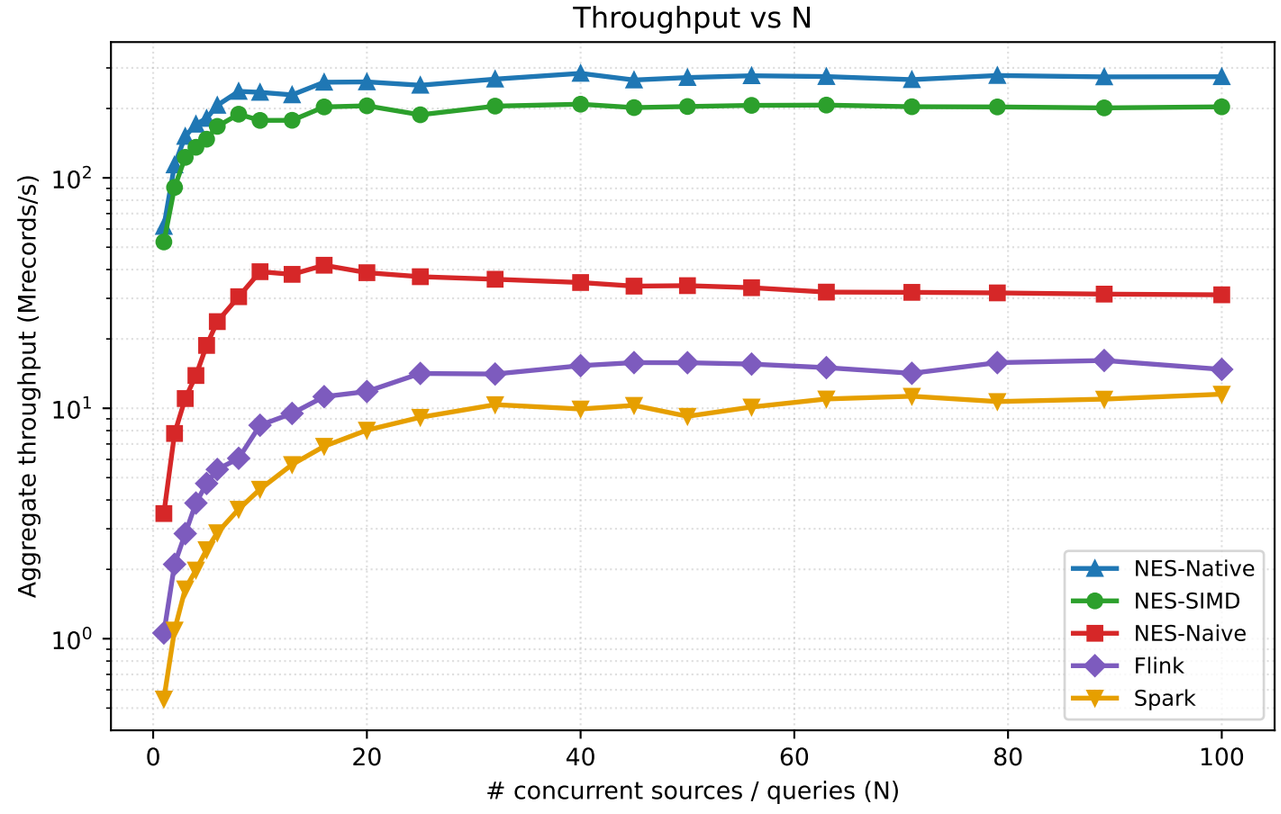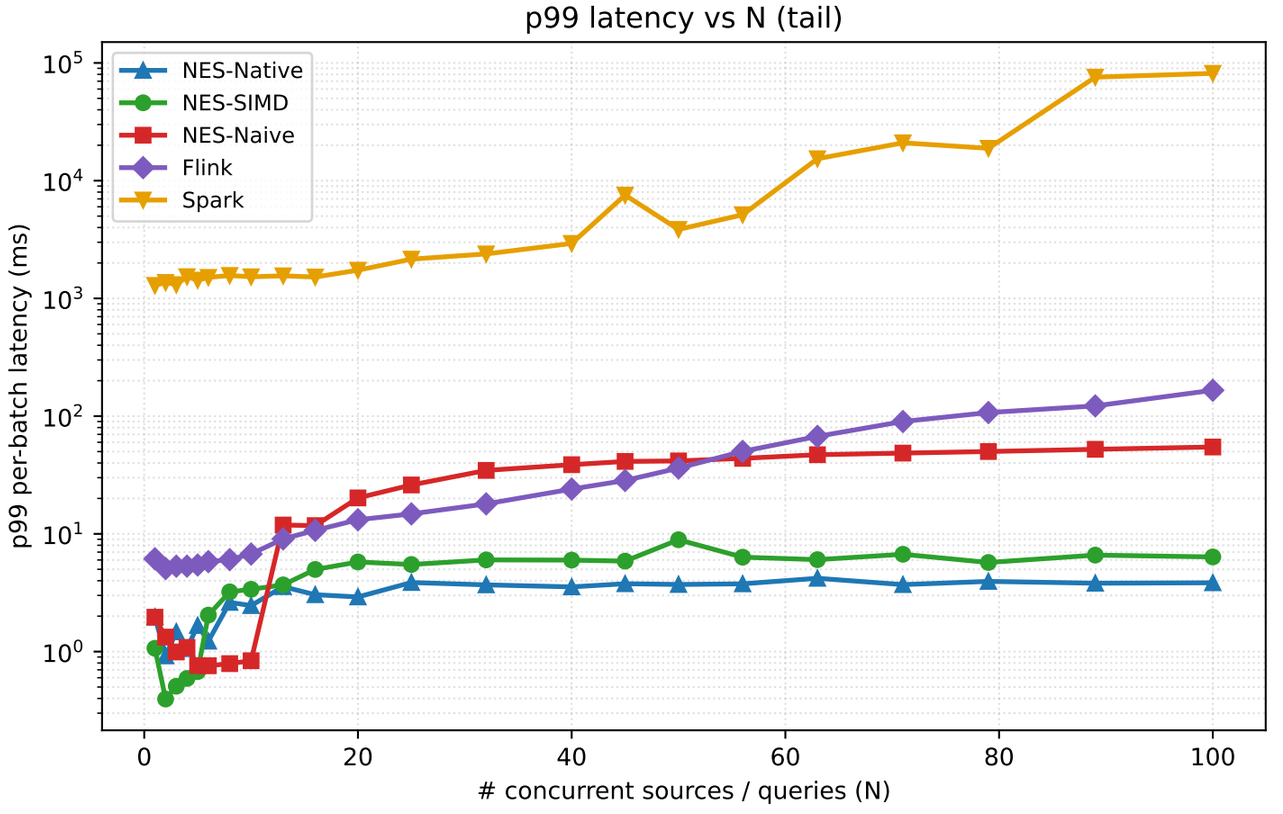Five-arm low-N scaling (N=1..100, 22-pt grid): Naive / SIMD / Native / Flink / Spark
10-query stateless mix, warm cache; throughput records/s (format-invariant)